

Experiment 4: 4. Outlier estimation and data cleaning using quartile and visualization methods.

Objective:

The objective of this experiment is to identify outliers in a dataset, estimate their impact, and perform data cleaning using quartile-based techniques and visualization methods.

Materials:

Dataset with potential outliers (e.g., CSV file, Excel spreadsheet)

Programming environment (Python, R, etc.)

Data analysis libraries (Pandas, NumPy, Matplotlib, Seaborn)

Procedure:

Data Preparation:

Import the dataset into the chosen programming environment. Familiarize yourself with the dataset's structure, features, and potential outliers.

Exploratory Data Analysis (EDA):

Conduct initial EDA to visualize the distribution of data. Create histograms and box plots to identify any skewness or presence of outliers.

Quartile-Based Outlier Detection:

Implement the quartile-based method to detect outliers:

Calculate the interquartile range (IQR) for each feature.

Identify lower and upper bounds using $Q1 - 1.5 * IQR$ and $Q3 + 1.5 * IQR$.

Flag data points falling outside these bounds as potential outliers.

Visualization of Outliers:

Use scatter plots and box plots to visually identify the potential outliers marked in the previous step. Highlight these points using distinctive colors or markers.

Impact Assessment:

Analyze the potential impact of outliers on data analysis or modeling tasks. For example, assess the effect of outliers on mean, median, and standard deviation.

Outlier Handling Strategies:

Based on the analysis, decide on appropriate strategies for outlier handling:

Remove outliers:

Delete the flagged data points that are deemed as outliers.

Impute outliers: Replace outliers with a reasonable value based on domain knowledge or statistical methods.

Data Cleaning:

Apply the chosen outlier handling strategy to the dataset and create a cleaned version. Document the changes made to the dataset.

Visualization of Cleaned Data:

Visualize the cleaned data using histograms and box plots to observe the impact of outlier removal/imputation on the distribution.

Comparison:

Compare the statistical measures of the original dataset with the cleaned dataset to understand the effect of outlier handling on data characteristics.

Documentation and Reporting:

Summarize the findings, including the identified outliers, the chosen handling strategy, and the impact on data quality. Provide insights into the importance of outlier detection and data cleaning in ensuring accurate and reliable analysis.

Results and Conclusion:

This experiment showcases the significance of outlier detection and data cleaning using quartile-based techniques and visualization methods. By identifying and addressing outliers, data quality is improved, leading to more accurate insights and robust analysis. The experiment emphasizes the importance of understanding the dataset's context and employing appropriate strategies for handling outliers.